

Government Engineering College, Rajkot
3rd Semester - (Odd 2026-27)
Data Structure (BE03000081)
Assignments
PBL Activity - III

Dead Line : 10/12/2026

Assignment - I

1. Explain Stack operations with algorithms.
2. Explain classification of data types.
3. Explain Insert, Delete operation in Queue with example.
4. Write algorithms to insert and delete elements from Singly Linked List.
5. Difference between Simple Queue and Circular Queue

Assignment - II

1. Construct binary search tree for the following data and find its inorder, preorder and postorder traversal - 10,3,15,22,6,45,65,23,78,34,5
2. Construct a tree for the given inorder and postorder traversals
Inorder DGBAHEICF
Postorder GDBHIEFCA..
3. What is an AVL tree? Explain the different types of rotations used to create an AVL tree with suitable examples.
4. Explain and write algorithm for Breadth First Search and Depth First Search with example.
5. Explain Spanning Tree algorithms.

Assignment - III

1. Write and explain Insertion Sort and Selection Sort algorithm.
2. Write and explain Merge Sort algorithm and Quick Sort algorithm
3. Write and explain Binary Search and Sequential Search Algorithms.
4. What is hashing? Explain Hashing Function.
5. Explain Collision-Resolution Technique.
6. Explain different file organization structure.